

Does Demographic Aging and Financial Globalization Moderate Crowding Out? Federal
Deficits and U.S. Long-Term Real Interest Rates.

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Honors Thesis

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Abstract

This thesis explores how U.S. federal budget deficits and demographic aging relate to long-term U.S. real interest rates, and how the role of global financial conditions may shape this relationship over time. Using annual data spanning from 1960-2023, I construct long-term real rates measure by subtracting 10-year treasury yields from CPI inflation, fiscal deficits as a percentage of GDP (gross domestic product) and demographic aging using the old-age dependency ratio. This study is motivated by an open economy loanable funds perspective in which government borrowing competes for savings, while the extent to which capital markets are globally integrated may influence how deficit borrowing relates to domestic real borrowing costs.

I estimate OLS regressions that include an interaction term between deficits and aging to see whether the marginal association between deficits and real rates varies with demographic structure. To reflect structural differences in the U.S. macroeconomy and extent of cross-border financial market integration over time, I report full sample estimates and compare results across two periods: 1960-1990 and 1991-2023. In addition to the baseline regression models, the later period models include a proxy for foreign real rates and net capital inflows, among others. I also ensure that standard errors are reported for all these models. Overall, the coefficients from the regressions suggest that the interaction of deficits and real rates matter, older demographic structures seem to mask at least partially the crowding out effect and the same can be said for capital inflows from the rest of the world. These results are consistent with the view that demographic structure and globally integrated capital markets jointly shape how deficit financing impacts the American bond market.

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I Introduction

The long-term fiscal sustainability of the U.S. federal government has come under heightened scrutiny as of late. The federal government has run a budget deficit every year since 2002, with a trend towards larger and more persistent budget deficits. Concerns surrounding the trajectory of federal debt are reflected in recent sovereign credit-rating downgrades by S&P (2011), Fitch (2023) and Moody's (2025) (Krauskopf 2025).

These fiscal concerns coincide with a significant demographic transition. As the American population ages and the baby-boom cohort progresses from late-career into retirement, the finances of the American state face headwinds: taxable income growth may slow while age-related spending (e.g. Medicare and related programs) is set to rise. While this channel points to demographics leading to increased fiscal pressure, the actual effect of aging on real long-term interest rates is theoretically ambiguous especially in an open economy. This is the case as a large cohort's retirement can lower the national savings rate while at the same time raising demand for safe assets as they prepare their portfolios for retirement, which could potentially place downward pressure on real rates to counteract the reduced savings channel.

This thesis examines how deficit-financed federal borrowing affects long-term real interest rates, while considering two opposing forces: (I) crowding out, which tends to raise real interest rates as public borrowing increases, and (II) population aging and financial globalization, which can increase demand for safe assets and thereby counteract crowding out. Understanding these interactions is important in assessing whether, and to what extent, federal borrowing displaces private sector activity. The thesis first reviews the macroeconomic theory underlying

crowding out and provides an overview of the relevant literature. It then presents econometric models that estimate the joint effects of deficits and demographic structure on real U.S. long-term rates (including an interaction term to allow the deficit effect to vary with age).

II Theory and Research Questions

To understand how fiscal deficits and demographic structure relate to real long-term rates, it is helpful to briefly review the macroeconomic frameworks connecting public borrowing, national savings and investment decisions, and cross-border capital movement in the open economy. This section summarizes these mechanisms and states the research questions that will guide the literature review and empirical analysis.

A standard starting point for understanding how fiscal policy and external balances relate to real rates is the national-income identity, which relates total output to spending components: $Y = C + I + G + NX$, with C being consumption, I for investment, G for government purchases of goods and services, and NX is net exports. It is important to note that government purchases are not the same thing as government spending; this is because government spending that goes to transfer payments (e.g. Social Security, SNAP, WIC, etc.) are not counted in G directly. Instead, transfer payments serve to increase households' disposable income which affects aggregate demand by increasing consumption C . In the short run, prices and wages are sticky, meaning that nominal variables (e.g. quantity of money in circulation) can affect real variables such as output and employment. This identity is useful here because it highlights the accounting links among government purchases, private spending and net exports—channels that show up later in the crowding out and capital inflow mechanisms.

John Maynard Keynes further extends this reasoning by modeling two markets. The liquidity preference-money supply (LM) relationship refers to the public's demand for liquidity

given the money supply; in equilibrium, it helps determine the short-term nominal interest rate for a given level of income (when subtracted from inflation is how one yields real interest rates as described by the Fisher equation). Longer term yields (e.g. 10-year treasury yields) can be seen as reflecting short-term yields plus a term premium. The investment savings (IS) relationship refers to equilibrium in the goods market; in equilibrium, it determines total output consistent with planned spending. A useful complementary line of thinking to this is the loanable-funds market, in which real interest rates act as the price that equilibrates saving and investment. This serves to influence the amount of capital formation in the economy as investment is usually a decreasing function of real rates: when rates rise, private investment should be expected to fall, all else being equal because the higher discount rates reduce the net present value of proposed capital expenditures, making such ventures less attractive to the management of firms.

Under IS-LM, an increase in the federal government's budget deficit (such as through higher spending or tax cuts) shifts aggregate demand upwards and increases short-run economic output. As income rises, money demand increases, putting upward pressure on interest rates. Higher interest rates tend to reduce interest sensitive private investments so some of the initial economic expansion is offset. This is sometimes referred to as crowding out, and persistent crowding out of investment could potentially reduce long-run capital accumulation.

Robert Mundell and J. Marcus Fleming further expanded this model to create the Mundell-Fleming model, also known as the IS-LM-FX model. This model preserves both IS & LM from Keynes's theories but adds FX (short for FOREX, or global currency markets into the model). This relates the money market to the goods market by looking at trade. If real rates rise

then domestic assets become more attractive to foreign investors to buy, causing the currency to appreciate, this will make the trade balance worse as domestic consumers will buy more foreign goods with their more valuable currency and foreigners will buy fewer domestic goods with their now less valuable currency.

IS-LM-FX predicts the same thing (for an open economy) and throws something else into consideration. As real rates rise due to the crowding out mechanism, foreigners are attracted to domestic investments. To buy domestic investments they must buy the local currency that the assets are denominated in; this buying raises demand for the domestic currency leading to its appreciation. This appreciation in turn deteriorates the trade balance via the mechanism outlined previously. This can reduce net exports and dampen effect of fiscal expansion on output relative to the initial increase in government demand.

Interestingly, when one observes historical interest rates in the American economy since the start of this century, one has difficulty observing the crowding out effect. The US persistently ran large budget deficits under the Bush Jr., Obama, Biden, and first Trump administrations, yet real rates trended downward for much of this time. This pattern seems consistent with scenarios in which global capital inflows, desire for safe assets, and demographics increases desired saving putting downward pressure on real rates.

The objective for this thesis is to consider the relationship between federal budget deficits, demographic structure, and real interest rates in the United States. Specifically, it investigates the following:

1. How do federal budget deficits affect real interest rates in the U.S.?

2. What is the role of demographic change, especially aging, in shaping real interest rate trends?
3. How do deficits and demographics interact with each other in capital markets?
4. Has financial globalization played a role in these developments?

III Review of Literature

III.1 Theoretical Foundations: Neoclassical v. Ricardian Equivalence

Significant effort in the economics profession has gone into refining the theoretical models that will then be applied by practitioners to explain real-world events. In previous times, one of the most contested questions in the literature regarded Ricardian equivalence. Ricardian equivalence argues that the federal government is bound by an *intertemporal* budget constraint, meaning that its ability to borrow from finance deficit spending is constrained throughout time. Meaning that if deficit-financed spending occurs today, the federal government must ultimately finance it through higher future taxes, lower future spending, or some combination of these fiscal adjustments, implying a burden likely to be borne by a future generation of taxpayers. Equivalence argues that households are rational economic actors whose utility is maximized by *intertemporal* consumption smoothing; given that tax cuts today must result in tax increases in the future, they will increase savings today to pay back the higher future tax bill to ensure consumption is as smooth as possible throughout the entirety of their lives (Kormendi 1983). Therefore, public finances do not impact aggregate demand or interest rates.

Alas for the proponents of equivalence, *homo economicus* does not exist. This theory makes several unrealistic assumptions, such as perfect ability to save and borrow throughout time to maintain stable consumption, complete ability to foresee the future and not discounting

future consumption by those in the present and knowledge about the existence of an intertemporal public finance constraint in the first place, the notion that most ordinary Americans will act this way is unrealistic at best (Modigliani and Sterling 1986). Gramlich (1989) finds that the Reagan-era deficits significantly reduced national saving, consistent with traditional crowding-out concerns. Graham (1995) critiques Kormendi's methodology directly, arguing that his treatment of income sources and fiscal variables creates simultaneity bias; when disaggregated correctly, Graham finds that government debt does, in fact, affect private consumption. More recent work by Gale and Orszag (2004) further undermines Ricardian assumptions. They show that a \$1 increase in federal borrowing reduces national saving by \$0.50 to \$0.80 and estimate that a 1% increase in the federal government's unified deficit raises long-term interest rates by 25 to 35 basis points, with even larger effects for the primary deficit. Taken together, these findings show that Ricardian equivalence is a questionable proposition at best, and that, in practice, deficits tend to reduce national saving and raise interest rates, supporting the neoclassical crowding-out hypothesis.

III.2 Empirical Evidence regarding Deficits & Real Rates

Crowding out in capital markets has long been a concern recognized among experts in the field; Indeed, the literature regarding this topic goes back decades. Crowding out can be observed over the long term when looking at real interest rates, but not in the short term. Meaning that short-term rates on instruments like bank CDs, commercial paper or T-bills are relatively unresponsive to immediate deficit increases. However, this does not mean that deficit increases have no impact, the long-term rate increases that it causes reduces capital accumulation in the private sector as capital budgeting projects are based on long-term rates as opposed to short-term rates when calculating the present value of the proposed capital expenditure projects;

the implication of this is that deficits lead to changes in real resource allocations in the American economy (Saunders 1989).

Research done during periods of higher deficits driven by exogenous political considerations concurs with the predictions made by theoretical macroeconomic models. A salient historical case illustrating the link between fiscal policy, real interest rates, and external imbalances is the Reagan administration's fiscal expansion in the early 1980s. In response to economic stagnation and geopolitical pressures, the administration enacted significant tax cuts and a major peacetime military buildup. These policies drove the federal budget deficit from 2.7% of GDP in 1980 to a peak of 6.1% in 1983. Although the deficit declined modestly thereafter, it remained well above pre-Reagan levels (Clinton White House Archives). As predicted by neoclassical crowding-out theory, the surge in public borrowing pushed up real interest rates, which in turn attracted substantial foreign capital inflows. This capital demand led to a sharp appreciation of the U.S. dollar—over 40% relative to major trading partners—which deteriorated the trade balance and contributed to a current account deficit approaching 3% of GDP by the mid-1980s (Chen, James; World Bank). Janet Yellen (1989) provides empirical support for this interpretation, arguing that the combination of deficit-financed tax cuts and defense spending raised real interest rates and depressed national saving, undermining the Ricardian view that deficits are neutral. Her analysis affirms the neoclassical perspective that fiscal deficits can crowd out investment and distort external balances, particularly when domestic saving fails to adjust in response to rising public debt.

The U.S. has seen a reduction in its external wealth in recent decades, the same decades in which large federal deficits were occurring. Private savings increases by forty-three cents for

every dollar increase in the federal deficit: foreign net investment increases by twenty-four cents producing a net reduction in private investment of thirty-three cents. Previous estimates showed that average long-run interest rates increase by two and a half basis points for every one percentage point increase in the national debt as a share of GDP, though this may have declined by two basis points; this is known as debt sensitivity of interest rates and its recent modest decline may be caused by a declining spread between treasury yields and the marginal productivity of capital (Congressional Budget Office, 2025). A glut of global savings and the demographic effects of the boomers may be causing the compression of the spread as the boomer's buying of treasuries in old age (as their retirement portfolios shift out of equities and into fixed income) puts downward pressure on treasuries and not on MPK. This modest recently observed phenomenon may mute crowding out at least temporarily. Similar estimates by other researchers have been obtained which appear to validate these estimates. Engen & Hubbard's paper *Federal Government Debts and Interest Rates* (2004), finds a 3-basis point increase in long-term interest rate per percentage point increase in the federal government's debt. This study was conducted before the CBO's more recent updated model and that fact that Engen & Hubbard's estimate of 3 basis points is similar to the CBO's older estimate of 2.5 basis points and gives one greater confidence in the econometric modeling of both papers.

However, several strands of the literature argue that crowding out can be muted under certain conditions. If the budget deficit is occurring during a recessionary gap, then crowding out cannot occur as the private sector is operating below its long-run capacity. Even in cases where the economy is at or near full employment deficits that are used to finance investments in public works (such as highways, rural electrification and hospitals, and R&D among others) may facilitate private sector investment to serve the demand stimulated by the public works. Public

works can encourage private investment by enabling private sector activities to be more profitable than they otherwise would be. An example of this would be a highway expansion resulting in a developer building a new truck stop. Therefore, breaking deficits into those that drive greater investment versus consumption needs to take place before a more complete analysis can be made (Eisner 1989). Additionally, bond financing may not be as damaging as is commonly believed, particularly to close recessionary gaps. Significant time lag makes causal analysis more challenging. If the government is to finance a deficit, the potential negative impact can be reduced by issuing short-term bonds instead of long-term bonds, so long-term private sector capital projects are not impacted. It should be noted that treasury bonds and private investment instruments are not perfect substitutes for each other, so a dollar-for-dollar “trade-off” of one instrument for another may not take place (Friedman 1978).

One should consider the possibility that factors outside of fiscal policy and underlying demographic structures may be partially driving some of these developments. Monetary policy can also have significant explanatory power in explaining a decades-long trend of declining real rates. A paper titled *Monetary Policy and the Secular Decline in Long-Term Interest Rates: A Global Perspective* by Hofmann, Li, and Wu (2025) attributes nearly 70% of that decline to the three-day windows surrounding U.S. Federal Open Market Committee (FOMC) announcements. The authors find that U.S. monetary policy shocks, not information effects or inflation expectations, are the primary drivers of this trend, with changes in expected short-term interest rates (risk-neutral rates) accounting for most of the decline. In contrast, monetary policy announcements from other central banks had minimal or no lasting impact on long-term rates, either domestically or globally. The study also shows that these effects extend to real interest rates, reinforcing the dominant role of U.S. monetary policy in shaping global bond markets. The

findings underscore the powerful international spillovers of U.S. monetary policy and suggest that the secular decline in global long-term interest rates is largely a monetary phenomenon, rather than purely structural or demographic. The centrality of the United States in setting the tone for global financial markets is also argued by others who assert that other countries have limited ability to influence capital markets on account of the enormous size of the American economy relative to the rest of the world and the pre-eminent role of the dollar as the reserve currency (Rey 2018).

III.3 Demographic Influences on Capital Markets

One major explanation for why real interest rates slowly declined throughout the first two decades of the twenty-first century is the country's underlying demographic change. This pattern is difficult to reconcile with what the standard theoretical models would predict. The key trends during this period include the influence of the baby boomers (largest generation in U.S. history), declining fertility rates after this cohort was born, and gradually increasing life expectancy (Carvalho, Ferrero, and Nechio 2017). Life expectancy has been increasing faster than retirement ages, resulting in longer retirements than for previous generations; baby boomers have responded to this by saving more for retirement than previous generations who had shorter expected life expectancy, this extra saving was concentrated during the prime of their working lives which occurred during the early twenty-first century making a large amount of capital available to the market for loanable funds. Other estimates also support this, with one study showing real rates falling by around 125 basis points due to higher savings for retirement by individual workers and a one-time increase in the size of the labor force due to women's entry en masse (Gagnon, Johannsen, and Lopez-Salido 2016). Given the persistence of baby boomers and women workers these researchers use the term "new normal" to describe the low-interest rate environment.

Because of persistently low fertility rates in the US (common among peer nations as well) the share of the middle aged and older population has consistently grown whereas younger people's share of the population has been shrinking. This results in more aggregate savings as people are spending less money on raising children that were never born. However, many models fail to fully appreciate the ramifications of this. An Affine Term Structure Model that explicitly incorporates the ratio of middle aged (40-49) to younger (20-29), can be a powerful predictive tool (Favero and Gozluklu 2015). They show that incorporating demographics improves the fit of models that predict bond market performance, especially for longer duration securities. These models can demonstrate that middle-aged households tend to save more and allocate a larger share of their portfolios to bonds, increasing demand for fixed-income assets, and putting downward pressure on yields. By embedding demographic dynamics into the structure of interest rate modeling, the authors provide a framework that better reflects slow-moving, predictable shifts in the supply and demand for capital, reinforcing the link between population aging and declining real interest rates. This effect is most pronounced during the pre-retirement years, when middle-aged households are actively saving and allocating to fixed-income assets. The implications of what may occur as this cohort transitions into retirement, when asset decumulation begins, will be explored later in this literature review.

Given that US domestic demographic trends were modeled in the last paragraph, the next logical step is to discuss other industrialized countries. Germany offers a great case study to this end, as its drop in fertility was sudden, occurring mostly in the 1970s, and offering a sharp before and after contrast for analysis (Lu and Teulings 2017). This generated an unusually large pre-fertility crash cohort whose size and lack of dependents enabled them to push extra capital into the German financial system, which put downward pressure on real interest rates. It is important

to consider that this is not a new steady state, as once that large cohort retires, they will stop contributing new capital and upon their death their heirs may liquidate many of the fixed income instruments resulting in the downward pressure revising, lowering bond pricing, and raising rates. Before that happens, the large cohort still has some time to go. Interest rates will not bottom out and then start increasing until 2035 based on this model. It appears that the 2030s will be crucial to the saga of an aging world, in the US the insolvency of Social Security is projected to occur sometime in the early to mid-2030s, so this timeline does not appear to be isolated to Germany.

Further literature extends the analysis beyond a single country. Other researchers using an overlapping generations model discuss multiple countries in one model. An overlapping generations model (or OLG) assumes that an economy has people at various stages of their lives all living in it. They are net consumers early in their lives to pay for college, cars, and housing, before being net savers and middle age to prepare for retirement, and then they draw down their retirement savings in old age. As that middle-aged share grows (due to the young people's share shrinking because of reduced fertility rates), real rates will fall as more money is supplied to the market. This model was used to find that demography alone explains about three-fourths of the 210-basis point reduction in rates over the prior 45 years (Lisack, Sajedi, and Thwaites 2017). They project a further decline of downward rate pressure beyond the timeframe given by Lu and Teulings. Wealth deaccumulation will likely be gradual for the elderly due to longer life expectancies and desire to bequeath assets to heirs. Other sources also validate the gradual unwinding of assets and slow rate increase projections. The Congressional Budget Office's 2009 report *Will the Demand for Assets Fall When the Baby Boomers Retire* report cites the same

reasons as Lisack et al. do for gradual unwinding. They also believe that large capital inflows into the US will likely serve to ameliorate the effects of any boomer asset sell-offs.

In total, demographic structure plays a critical role in shaping capital market conditions over time. Countries with a large youth population tend to experience upward pressure on real interest rates due to strong demand for loanable funds. In contrast, economies dominated by middle-aged cohorts experience greater saving and capital accumulation, placing downward pressure on rates. As populations transition toward old age, gradual asset decumulation is likely to exert upward pressure once again, though this process may unfold slowly. As the Congressional Budget Office notes, global capital flows—particularly foreign demand for U.S. assets—may soften or delay these effects. These insights underscore the importance of integrating demographic forces into any analysis of interest rate behavior. The next section builds on this foundation by examining how open capital markets and international financial dynamics further shape the relationship between domestic fiscal policy and real interest rates.

III.4 Open Economy Considerations

Financial globalization has continued apace for decades, therefore understanding the open economy is essential. The literature review touched upon the open economy briefly in section III sub-section II *Empirical Evidence Regarding Deficits & Real Rates*. Therefore, brevity will be considered for this section; the twin deficits hypothesis was already explored with a case study from the US in the 1980s. Evidence shows that this also holds true in other countries. Panel data from eighty-nine countries between 1971 and 1995 show a strong correlation between budget surplus and current account surpluses (Chinn and Prasad 2000). This is the same thing as saying budget deficits lead to current account deficits. Having such a wealth of data from so many countries further supports the view that fiscal expansions reduce domestic

saving leading to lower external wealth. Further research indicates that fiscal consolidation (i.e., belt tightening such as tax increases and spending cuts) equal to 1% of GDP improves the current account balance by 0.5% of GDP (International Monetary Fund 2011).

The current account deficit increases in response to a budget deficit increase is because the financial account surplus is growing; this helps fund the deficit and keep domestic interest rates lower for the US than they otherwise would be. Additionally, the US's status as the global reserve currency ensures that global economies will be tied to the US. The centrality of the US is demonstrated by the fact that US monetary policy shocks generate global movements, even in countries with floating exchange rates (Miranda-Agrippino, Silvia, and Rey 2020). For this thesis, the findings support controlling for international capital flows, which may mute the effect of federal deficits on real rates by increasing foreign demand for U.S. debt.

IV Data & Methodology

IV.1 Data Overview

This study uses an annual U.S. time-series dataset assembled from aggregating easily accessible publicly available sources including Federal Reserve Economic Data (FRED), the World Bank, the Congressional Budget Office (CBO), the Bureau of Economic Analysis (BEA), and selected financial-market data sets. The dependent variable of interest is the real rate on 10-year U.S. treasury bonds (r_t^{US}), with nominal rates ($GS10_t$), and inflation (π_t).

$$r_t^{US} = GS10_t - \pi_t$$

The key explanatory variables are the federal budget deficit as a percentage of gross domestic product (GDP) and the old-age dependency ratio, a measure of old people as a percentage of the working age population. I have included multiple other control variables to

account for capital inflows from the rest of the world, business-cycle dynamics, among other macroeconomic variables.

The primary sample of interest runs from 1960-2023 (N=64), which reflects the intersection of the real-rate series with the old-age dependency ratios and key controls. In addition, I included a smaller balanced sample spanning 1983-2023 (N=41) which is useful for robustness checks that included a wider set of controls that begin in the early 1980s. All variables are expressed in percentages or percentage points/share of GDP to make the coefficients easier to interpret. Appendix Table A4 reports descriptive statistics (mean, standard deviation, minimum, and maximum) for the main regression variables used in the core-controls specification over 1960-2023.

IV.1.1 Variable Sources & Description Table

Table 1—Variable Definitions and Data Sources

Category	Stata name	Variable description	Construction notes / units	Source (series)
Dependent variable	real_rate_us	U.S. real 10-year interest rate	Annual real 10-year U.S. rate (ex-post). Constructed as 10-year nominal Treasury yield minus CPI inflation; annualized to calendar-year frequency.	Constructed: FRED GS10 + CPI inflation
Key regressors	deficit_pos	Federal budget deficit (% of GDP, + = deficit)	Constructed from deficit_gdp (FRED FYFSGDA188S). Transformed so higher values indicate larger deficits.	FRED (FYFSGDA188S)
Key regressors	old_dep_ratio	Old-age dependency ratio	Old-age dependency ratio: population 65+ divided by population ages 15-64 (percent).	World Bank: SP.POP.DPND.OL

IV.1.2 Variable Construction

because it is a large, advanced economy with a deep sovereign bond market and long-horizon data coverage; its yields help capture broader global-bond market conditions rather than just U.S.-specific fiscal dynamics. Because German reunification (1990) may introduce a structural break in German macro/financial series, the German real 10-year rate is used strictly as a proxy for global interest rate conditions and is not used for analysis beyond this role. Saving behavior (psav_rate), unemployment rate (u_act), and federal revenue collection, (receipts_gdp). International financing controls are measured using net equity, net debt and total net inflows as a share of GDP (Dallas Fed Economics chart data).

IV.2 Data Cleaning, Annualization and Alignment

All data series are converted to annual frequency and aligned by calendar year prior to being loaded into Stata for analysis, except for measures that are based on fiscal year (such as the deficit). Some of the series were originally reported as monthly figures, for these, annual values are calculated by taking the mean of the twelve-monthly observations in that calendar year. Annual averaging ensures that each series is measured consistently at the same frequency and to reduce short-term volatility's impact when conducting statistical analysis aimed at reflecting longer-running fiscal and demographic developments.

After each data series was annualized, a master spreadsheet was assembled by matching observations based on calendar year. Through this assembly process in Excel, the year serves to ensure that alignment is maintained throughout all the series, ensuring that each row of the master spreadsheet is the same year for all fiscal, demographic and financial variables. A separate working spreadsheet was used to calculate derived variables (e.g. cyclical unemployment, which was the actual unemployment rate – natural rate of unemployment), and

these constructed series were then copied over into the master spreadsheet. Upon full assembly of the master spreadsheet, it was imported into Stata to run the regressions.

After importation, the sign for deficits was flipped such that higher values indicate a larger deficit as a share of GDP, while negative values indicate surplus. This was done to make interpretation of coefficients more intuitive for the reader. The interaction term between deficits and demographics was also created inside of Stata as well.

Missing observations are handled by running each regression only on years with complete data for all the variables included in that specification. Therefore, the estimation sample differs between different regressions depending on the availability of included controls, as seen through reported observation counts. No imputation (including interpolation) was conducted; regressors are observed years only.

IV.2.1 Empirical Specification

The empirical analysis estimates the relationship between real interest rates on 10-year U.S. treasury bonds, federal government budget deficits, and demography using annual U.S. time series regressions. Ten-year treasuries are used as the benchmark long-term risk-free rate, so variation in yields is driven not be default risk. The baseline specification is as follows:

$$r_t = \beta_0 + \beta_1 d_t + \beta_2 a_t + \beta_3 (d_t \times a_t) + \varepsilon_t$$

To account for other determinates of long-term real rates, I use additional models that add a vector of controls:

$$r_t = \beta_0 + \beta_1 d_t + \beta_2 a_t + \beta_3 (d_t \times a_t) + \gamma' X_t + \varepsilon_t$$

where r_t is the real 10-yr. treasury rate (real_rate_us), d_t is the federal deficit as a percent of GDP (deficit_pos), a_t is the old-age dependency ratio (old_dependency_ratio), and $(d_t \times a_t)$ is

an interaction of the two, and X_t is a placeholder for additional control variables (e.g. saving behavior, labor market conditions, and net capital inflows), which are shown in the regression tables listed in the next chapter (tables 2-3)

As global financial integration and the international macroeconomic environment differ across historical periods, I also model pre-1991 and 1991-and-beyond subsamples. This split is intended to capture structural differences that may be present with the end of the Cold War and the subsequent era of deeper capital mobility, which has the potential to affect the reduced-form relationships that this thesis intends to study.

V Regression Results

V.1 Baseline model, domestic and global controls, and core-controls specification

Chapter 5 presents regression results on the relationship between real U.S. 10-year rates, federal budget deficits, and demographic aging. The model has an interaction term, where the marginal effect of deficits on real rates is $\frac{\partial r_t}{\partial d_t} = \beta_1 + \beta_3(d_t \times a_t)$. Observation counts differ across specifications due to differences in control-variable coverage. Table 2 presents the baseline model, baseline model with various controls and the core-controls specification (Table 2, column 4), which prioritizes the theoretically important controls while preserving sample coverage.

*Table 2. Real Rate Regression Results—Baseline, Controls, and Core-controls specification
(Dependent variable: U.S. real 10-year Treasury rate)*

Variable	(1) Baseline	(2) + Domestic macro controls	(3) + Global & capital controls	(4) Core-controls specification
Deficit (% GDP, + = deficit)	1.879*** (0.599)	0.955 (0.617)	1.598*** (0.580)	0.118 (0.447)
Old-age dependency ratio	0.010 (0.147)	-0.073 (0.211)	1.381*** (0.261)	0.827*** (0.184)
Deficit × Old-age dependency ratio	-0.090*** (0.028)	-0.056* (0.028)	-0.104*** (0.029)	-0.022 (0.018)
Term spread (10y– 3m, pp)	—	-0.219 (0.239)	—	—
Natural unemployment rate (%)	—	1.967 (1.225)	—	—
Personal saving rate (%)	—	0.437** (0.174)	0.457** (0.178)	0.008 (0.091)
German real 10y rate	—	—	1.200*** (0.150)	1.141*** (0.157)
Cyclical unemployment (%)	—	—	0.856*** (0.269)	—

Total net capital inflow (% GDP)	—	—	0.549** (0.228)	—
Unemployment rate (%)	—	—	—	0.561** (0.225)
Federal receipts (% of GDP)	—	—	—	-0.423* (0.251)
Constant	1.592 (2.588)	-8.062 (9.180)	-30.513*** (6.295)	-10.985** (5.112)
Observations	64	42	48	64
R-squared	0.2456	0.8384	0.8011	0.6276

Coefficients reported with standard errors in parentheses *p < 0.10 **p < 0.05 ***p < 0.01

From Table 2, in the baseline model deficits are associated with higher real rates, however when interacted with demographics the effect is conditional; the marginal effect of crowding out appears to weaken as the population ages. For column two, when domestic controls and saving are included in the model, the deficit becomes statistically insignificant in influencing real rates, the interaction term remains negative, implying the marginal deficit effect declines as the old-age dependency ratio rises. The term spread and natural unemployment rates are not significant on real rates and the personal savings rate (higher personal savings) is associated with higher real rates. While this result is seemingly counter-intuitive on its face, what we may be observing here is the direction of causality in determining household savings levels flows from real rates to behavior (in that people will save more when they can get a higher return); however, it is important to acknowledge that as column two has the smallest number of observations, these results may not be as robust as the other columns' outputs. For column three, deficits, old age and the interaction of the two are all statistically significant. German real rates, which this study

uses as a proxy for the risk-free rate in other countries exhibits strong co-movement with U.S. real rates and is statistically significant; given the increasingly globalized nature of international capital markets, this finding makes sense. Lastly in column four, which is the core-controls specification, global real-rate conditions (German proxy), labor market conditions, and aging remain statistically significant determinates of U.S. real rates, while the deficit and deficit-interaction terms are not statistically distinguishable from zero¹.

V.2 Additional Robustness and Diagnostics

As an added robustness check, a saturated specification including a broader set inclusive of all macro and fiscal controls is estimated and reported in Appendix Table A2. The saturated model is intended to evaluate whether the coefficient estimates in Table 2 hold when they are heavily controlled for; alas, coefficient-by-coefficient interpretation is limited by the smaller sample size and collinearity among related fiscal variables. Diagnostic checks for multicollinearity are summarized using variance inflation factors are provided in Appendix Table A1. And a selected correlation matrix for the core variables can be found in Appendix Table A3 for descriptive content.

¹ Additional one-at-a-time robustness checks leave the broader interpretation intact. In the pre-1991 sample, removing the interaction term yields a positive and statistically significant deficit coefficient (1.48***), whereas removing the German control leaves the main sign pattern largely unchanged. In the post-1990 sample, by contrast, the results seem more sensitive: the baseline model shows a positive and significant deficit coefficient (0.85**) and a negative and significant interaction term (-0.05**), but removing the interaction reduces the deficit coefficient to -0.04, and removing the German control changes the deficit coefficient to negative and insignificant (-0.34), while making the old-age dependency ratio negative and highly significant (-0.61***). Adding a recession dummy does not materially change the main results in either subsample, and adding a one-year lag is more informative in the later-period sample than the pre-1991 sample: in the pre-1991 sample, lagging the deficit leaves the deficit term positive but statistically insignificant, whereas in the post-1990 sample the lagged deficit coefficient remains positive but only marginally significant (0.48*) and the lagged interaction term remains negative and significant (-0.03**).

VI Pre v. 1991 and beyond results

Table three reports split estimates for the pre-1991 and 1991-beyond periods.

Columns (1)-(2) report pre-1991 specifications, while columns (3)-(4) report 1991 and beyond specifications; control columns add the same open-economy block.

Table 3. Real Rate Regression Results—Pre-1991 vs. Post-1990 Specifications (Dependent variable: U.S. real 10-year Treasury rate)

Variable	(1) Pre-1991: baseline	(2) Pre-1991: + global & inflows	(3) Post-1990: baseline	(4) Post-1990: + global & inflows
Deficit (% GDP, + = deficit)	-18.218*** (4.379)	10.676 (10.959)	-0.048 (0.540)	0.851** (0.401)
Old-age dependency ratio	-2.775*** (0.737)	4.233** (1.683)	-0.504** (0.224)	0.513* (0.270)
Deficit × Old-age dependency ratio	1.172*** (0.267)	-0.540 (0.638)	-0.002 (0.026)	-0.046** (0.021)
German real 10y rate	—	-0.244 (0.760)	—	0.945*** (0.218)
Total net capital inflow (% GDP)	—	1.243** (0.415)	—	-0.076 (0.163)
Personal saving rate (%)	—	1.675** (0.593)	—	0.123 (0.147)
Constant	44.467*** (11.384)	-90.543** (31.383)	12.034*** (4.256)	-10.109 (6.001)
Observations	31	15	33	33
R-squared	0.598	0.858	0.538	0.892

Coefficients reported with standard errors in parentheses *p < 0.10 **p < 0.05 ***p < 0.01

Note: German reunification (1990) may represent a structural break in German series; the German real rate is included only as a global-rate control.

As stated above, this table reports results from a split model pre-1991 and beyond, with the baseline model consisting of the deficit, old-age dependency ratio, and their interaction; the additional controls for the open-economy model include German real rates, total net capital inflows as a share of GDP, and the personal savings rate.

In the pre-1991 baseline specification (column 1), the deficit coefficient is negative and statistically significant, and the interaction of the two is positive and significant. Although the individual signs may appear counter-intuitive in isolation, it is important to keep in mind that the economic meaning of the output comes from interpreting the joint terms at the old-age dependency ratio at any given time. As deficits are interacting with aging, the implied association between deficits and real rates depends on the level of the old-age dependency ratio. Column 2, which is the pre-1991 baseline with global control variables. Given that data before 1991 are limited, the sample size is greatly reduced ($N=15$). With so few observations, coefficient instability is a significant concern as this regression is best viewed as a sensitivity check rather than definitive evidence of how capital markets operated pre-1991. In the post 1990 baseline (column 3), the deficit and interaction terms are close to zero and statistically insignificant, while aging is negatively associated with real rates.

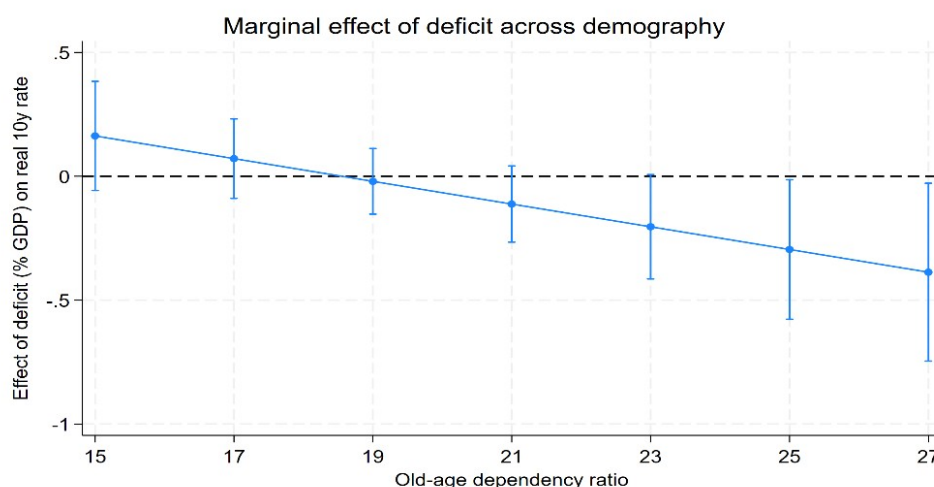
Using coefficients from table 3's baseline model, the marginal deficit effect is positive in twenty-five out of the 31 of the pre-1991 years, which is broadly consistent with crowding out as predicted conventional macroeconomic theory. By contrast, in the post-1990 global macro variables model, the marginal deficit effect is positive in only 10 of the 33 years, as described in further detail below.

Once the open economy is included in the regression output (column 4), the results have considerably more explanatory power (r-squared of .892 v. .538 for the column 3 baseline). First, German 10-year rates have a large and statistically significant coefficient with U.S. real rates (0.945), meaning that for every one percentage point increase in German real rates, American real rates rose by .945 percentage points (94.5 basis points), this finding makes sense given the globalized nature of contemporary cross-border capital markets. Second, the deficit coefficient becomes positive and statistically significant (0.851), while at the same time the deficit interaction term is negative and statistically significant (-0.046) indicating that the positive association between deficits and higher real rates becomes weaker as the old-age dependency ratio increases. Note that net capital inflows and domestic saving behavior are not statistically significant, indicating that the inclusion of these add limited explanatory power after global interest rates are already accounted for.

VI.1 Marginal Effects of Demography in the Contemporary Open Economy

As marginal effects of the budget deficit have an interaction term with aging, the post 1991 specification implies a marginal effect of $0.851 - 0.046 a_t$, which crosses zero at around $a_t \approx 18.5$. For comparison, at $a_t = 15$, the marginal effect is around 16.1 basis points, and at $a_t = 20$, its at around -7 basis points. Thus, the deficit's impact on capital markets depends on demographics. Figure 1, which was generated in Stata, provides a quick visual of this.

Figure 1—Marginal effect of deficits on U.S. real 10-year Treasury rates across the old-age dependency ratio (post-1990 open-economy specification)



Notes: Points show estimated marginal effects of the deficit-to-GDP ratio evaluated at selected values of the old-age dependency ratio; bars indicate 95% confidence intervals. The dashed horizontal line denotes zero marginal effect.

VII Summary of Results

Overall, these results show the relationship between U.S. fiscal deficits and 10-year rates in real terms on U.S. treasuries is regime-specific and sensitive to global financial conditions. When observing demographic values, the implied marginal deficit effect is positive in most of the pre-1991 years but near zero or negative in most post-1990 years. A plausible explanation for what may be happening here is that in later decades (1990s through the 2010s) the baby boomer cohort is nearing retirement, as they save these funds pool, generating what has been described as a “global savings glut,” which is why older demography in later years is associated with marginally lower real rates. Global interest rates may have also been suppressed during this time due to high savings rates in China and Japan and the growing sovereign wealth funds of the oil-exporting nations; all of whom supplied funds to global capital markets. Thus, these two sources of funds could have global interest rates low for a prolonged period, serving to mask the

crowding out effect, this claim is supported by the strong evidence for U.S.-German interest rate co-movement in the post-1990 global macro controls regression. In essence, the U.S. federal government has been able to borrow significantly more money, and on considerably better terms in more recent decades than likely would have been possible in the past.

VIII Shortcomings and Limitations

VIII.1 Data and Measurement Limitations

This study relies on an annual U.S. time-series dataset that was assembled by compiling data from several different sources into a single spreadsheet to ensure a consistent format to run the regressions. While this approach improves transparency and reproducibility, it introduced several limitations. First, the dependent variable was constructed as a real 10-year Treasury rate using nominal yields minus ex-post inflation, which may differ from a forward-looking real rate based on expected inflation. If inflation expectations are significantly divergent from realized inflation, the dependent variable may contain measurement error, especially in the era of high U.S. inflation during the 1970s and early 1980s. Second, several of the explanatory variables, especially fiscal and demographic measures, move slowly over time, increasing the risk that results reflect shared trends rather than short-run causal effects. Finally, the use of annual averages may smooth over meaningful intra-year fluctuations, which may be important during a recession or periods of swift policy changes.

VIII.1.1 Omitted Mechanisms and Institutional Channels

A separate limitation is that these regressions may omit institutional and structural factors that can influence the demand for treasury securities (which this study uses for the real-long term rate). For instance, changes in financial regulations can increase structural demand for safe assets such as treasuries. If commercial banks and related financial institutions face stronger liquidity or

capital-related incentives under different regulatory environments to hold treasuries, this can place downward pressure on treasury yields independent of deficits, demographics, and their interactions. Additionally, slow moving fundamentals such as trend productivity growth may have influenced real rates over time and may be correlated with demographic aging and fiscal variables. Estimates of total factor productivity and long-run economic growth affect the desire of companies to tap capital markets to finance capital expenditures. If companies en masse believe future growth to be strong, they may borrow more, placing upward pressure on real rates, and the reverse may also hold. This may have occurred in more recent times, if future economic growth is going to be slower because of slower increase in the size of the labor force, then they may pull back on capital expenditures that otherwise would have taken place. In this event, omitted-variable bias could affect estimated deficit and interaction coefficients in reduced-form time series regressions.

VIII.1.2 Identification and Endogeneity Concerns

Time series macroeconomic regressions also present inherent identification challenges.

The regressions in this thesis are reduced form, in that they estimate conditional associations describing how real long-term rates move with federal deficit and demographics (aging), holding included controls constant. This differs from estimating the causal effect of an exogenous fiscal-policy shock. Fiscal outcomes respond endogenously to business-cycle conditions that also influence interest rates. For example, recessions widen deficits through automatic stabilizers and can simultaneously affect long-term rates through monetary policy responses and changes in risk sentiment (e.g. flight to safe assets, such as treasuries). Although several regressions include controls for macroeconomic variables, these controls do not fully eliminate simultaneity between fiscal variables and interest rates. In addition, reverse causality is also a concern. Higher interest

rates can mechanically interest federal net interest outlays, thereby leading to larger budget deficits, even absent crowding out effects. For these reasons, the coefficient estimates should be interpreted as reduced-form relationships rather than definitive causal estimates.

VIII.1.3 Sample size and regime split constraints

The analysis is further constrained by sample size and coverage. Even the full sample only includes sixty-four annual observations, and adding controls reduces the effective sample due to data availability. The pre-1991 open-economy regression is especially limited (N=15), making coefficient instability and sensitivity to outliers a non-trivial concern. Splitting the sample into older and modern periods helps improve interpretability when considering macroeconomic regime differences, but at the cost of reducing statistical power and makes inferences more sensitive to the inclusion of added controls. Therefore, split-sample outputs should be viewed as suggestive evidence for structural changes rather than definitive causal evidence of regime differences.

VIII.1.4 Multicollinearity and specification sensitivity

Several key regressors, including deficits, demographics, fiscal receipts, and the savings rate is correlated and trend with one another over time. This creates concerns around multicollinearity and inflated standard errors. Variance inflation factors reported in the appendix confirm that multicollinearity is a meaningful issue, particularly for the deficit-demography interaction term. Though it should be noted that this limitation is not unusual for long-term macro time series work. The most reliable interpretation focuses on robust patterns between the models, such as the strong co-movement of U.S. and foreign (German) real rates in the 1991-

beyond period and the conditional nature of deficit effects implied by the interaction structure, rather than on any single coefficient in isolation.

IX Conclusion

This thesis looks at how U.S. federal budget deficits and demographic aging relate to real long-term interest rates, using an annual time series from 1960-2023. These regressions run on this time series emphasize an interaction between fiscal policy and the old-age dependency ratio, allowing for the implied relationship between deficits and real rates to vary with the level of the old-age dependency ratio. As macroeconomic time-series relationships can be sensitive to global conditions and policy regimes, regressions are run that incorporate both controls for domestic and global macroeconomic variables.

The results provide evidence that the deficit-real-rate relationship is regime-specific and sensitive to the global macroeconomic environment. In the full-sample regressions, deficits are associated with higher real long-term rates when running the baseline model, which is consistent with crowding out theory from contemporary macroeconomic theory. However, this relationship becomes weaker and less precisely estimated upon the inclusion of additional controls. This can be seen in the core-controls specification which addresses control variables related to global capital markets and the domestic labor market. Across models inclusive of the foreign rate proxy, U.S. and German interest rates feature strong co-movement, which makes sense in light of the increasingly globalized nature of capital markets; domestic rates are strongly influenced by foreign developments to a greater degree than in the earlier decades in this study.

The split-sample regressions further suggest that the fiscal-real rate relationship differs across these two eras. In the pre-1991 period, implied marginal deficit effects are more

frequently positive when examined at observed demographic values, which is broadly consistent with crowding out being easier to see in a less financially integrated world, one in which domestic developments play a bigger role in capital markets. In the post-1990 period, estimates show that the marginal deficit effect is frequently small or even negative when global variables are controlled for, and the interaction term implies that the overall association between deficits and demographics weakens as the population ages. One can refer to the figure included in the section about the marginal effect in more detail.

These findings should be interpreted as reduced-form relationships rather than definitive causal estimates of exogenous fiscal shocks. Nonetheless, the regression outputs suggest that the post-1990 figures are best interpreted as conditional on the global macro-financial environment of that period. If the major sources of saving and capital inflows from abroad were to weaken (such as from demographic transitions in surplus countries or from reduced revenues from oil sales), it is within the realm of possibility that the sensitivity of real rates to federal deficits could re-emerge.

In summary, this thesis finds that the relationship between federal deficits and long-term real rates is not constant over time and appears to be regime dependent. In the pre-1991 era, the estimated marginal deficit effects are more often positive at observed demographic values, consistent with crowding-out being more visible in smaller, less financially integrated capital markets. In the post-1990 era, the marginal deficit effect is frequently small or negative once controls are introduced, and the interaction results suggest that the association weakens as the population ages. Overall, these results indicate that crowding out can be muted by broader global

conditions and demographic dynamics rather than disappearing as a permanent structural feature of American bond markets.

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XI Data Sources Used

U.S. Congressional Budget Office (CBO). Historical fiscal series used to construct receipts (% of GDP) and spending subcomponents (% of GDP), including mandatory outlays, discretionary outlays, and net interest (% of GDP).

Federal Reserve Bank of Dallas. Dallas Fed Economics chart data used to construct net capital inflows as % of GDP (net equity inflows, net debt inflows, and total net inflows).

Federal Reserve Bank of St. Louis (FRED). U.S. Treasury yields and spreads (e.g., GS10; T10Y3M), labor-market series (e.g., UNRATE; NROU), Federal Reserve balance sheet series (e.g., WALCL; WSECOUT), and other macro/financial indicators accessed via FRED.

Kenneth R. French Data Library. Equity-market series used in construction of financial controls (e.g., equity risk measures depending on specification).

OECD (via FRED). Main Economic Indicators long-term government bond yield series for Germany (10-year “main/benchmark” yield), used to construct the German long-term real rate control.

Our World in Data (OWID). Market capitalization of listed domestic companies (% of GDP) series used as a financial-market control (market cap / GDP).

U.S. Bureau of Economic Analysis (BEA). National accounts and/or international transactions data used for investment (% of GDP) and related macro controls (where sourced directly from BEA).

U.S. Inflation Calculator. CPI inflation series used to construct ex-post real 10-year rates from nominal yields.

World Bank. World Development Indicators (WDI) series including demographic ratios (old-age dependency ratio) and other macro series used as controls (including series accessed via FRED where applicable).

YCharts. U.S. investment (% of GDP) series used as an investment control (kitchen-sink specification).

XII Appendix

This appendix reports supplemental diagnostics and robustness checks referenced in chapter 5.

Table A1 Variance Inflation Factors (VIFs)

Panel A. U.S. Real 10-year Treasury Rates—Core-controls specification (Table 2, column 4)

Variable	VIF	Tolerance
Deficit (% GDP, + = deficit)	106.777	0.009
Old-age dependency ratio	10.078	0.099
Deficit × Old-age dependency ratio	107.610	0.009
German real 10y rate	3.528	0.283
Unemployment rate (%)	3.715	0.269
Personal saving rate (%)	2.386	0.419
Federal receipts (% of GDP)	2.054	0.487
Mean VIF	33.736	—

Notes: VIFs computed for the core-controls specification (Table 2, column 4). High VIFs are expected for the interaction term and its components because they are mechanically correlated.

Panel B. U.S. Real 10-year Treasury Rates—Saturated (“kitchen-sink”) specification (Appendix Table A2)

Variable	VIF	Tolerance
Deficit (% GDP, + = deficit)	1,445.217	0.001
Old-age dependency ratio	113.851	0.009
Deficit × Old-age dependency ratio	1,449.401	0.001
Federal receipts (% of GDP)	13.852	0.072
German real 10y rate	18.409	0.054
Net equity inflow (% GDP)	6.632	0.151
Net debt inflow (% GDP)	13.147	0.076
Open market operations (% GDP)	7.804	0.128
U.S. investment (% of GDP)	8.568	0.117
U.S. public debt (% of GDP)	76.709	0.013
Discretionary outlays (% of GDP)	85.639	0.012
Net interest (% of GDP)	23.636	0.042
Personal saving rate (%)	14.286	0.070
Natural unemployment rate (%)	245.423	0.004
Unemployment rate (%)	18.677	0.054
Term spread (10y–3m, pp)	4.875	0.205
Baa risk-free spread (pp)	4.205	0.238
Market cap (% of GDP)	58.562	0.017
Equity risk premium (pp)	5.286	0.189
Mean VIF	190.220	—

Notes: Very large VIFs are expected in this setting because several variables trend together over time, and the deficit term and its interaction with aging are mechanically correlated with their components.

Panel C. U.S. Real 10-year Treasury Rates Post-1990 marginal-effects specification (Table 3, column 4)

Variable	VIF	Tolerance
Deficit (% GDP, + = deficit)	121.541	0.008
Old-age dependency ratio	5.910	0.169
Deficit × Old-age dependency ratio	153.031	0.007
Term spread (10y–3m, pp)	2.408	0.415
Natural unemployment rate (%)	4.840	0.207

Personal saving rate (%)	3.293	0.304
Mean VIF	48.504	—

Notes: Because the main quantity of interest is the deficit marginal effect, which is dependent on aging, the presence of high VIFs provides a rationale for emphasizing the implied marginal effects (Figure 1) rather than interpreting each individual coefficient in isolation.

Table A2 U.S. real 10-year Treasury rates kitchen-sink regression

Variable	Coefficient	Robust SE
Deficit (% GDP, + = deficit)	-0.041	1.287
Old-age dependency ratio	1.542***	0.450
Deficit × Old-age dependency ratio	-0.024	0.055
Federal receipts (% of GDP)	-0.492	0.552
German real 10y rate	0.564***	0.188
Net equity inflow (% GDP)	0.047	0.270
Net debt inflow (% GDP)	-0.328	0.242
Open market operations (% GDP)	0.120	0.111
U.S. investment (% of GDP)	0.654***	0.222
U.S. public debt (% of GDP)	-0.113**	0.049
Discretionary outlays (% of GDP)	0.063	0.833
Net interest (% of GDP)	-0.077	0.756
Personal saving rate (%)	0.049	0.189
Natural unemployment rate (%)	0.028	3.250
Unemployment rate (%)	1.226***	0.239
Term spread (10y–3m, pp)	0.204	0.224
Baa risk-free spread (pp)	-0.364	0.320
Market cap (% of GDP)	-0.009	0.020
Equity risk premium (pp)	0.025	0.017
Constant	-29.810**	13.400
Observations	41	
R-squared	0.953	

Notes: OLS estimates for the saturated (“kitchen-sink”) specification. Robust standard errors reported in the third column. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A3 Pairwise correlations (core variables)

Variable 1	Variable 2	Correlation
Deficit (% GDP, + = deficit)	U.S. real 10y rate	-0.296
Old-age dependency ratio	U.S. real 10y rate	-0.734
German real 10y rate	U.S. real 10y rate	0.880
Unemployment rate (%)	U.S. real 10y rate	0.333
Personal saving rate (%)	U.S. real 10y rate	0.325
Federal receipts (% of GDP)	U.S. real 10y rate	0.021
Deficit (% GDP, + = deficit)	Old-age dependency ratio	0.458
Deficit (% GDP, + = deficit)	German real 10y rate	-0.394
Deficit (% GDP, + = deficit)	Unemployment rate (%)	0.583
Deficit (% GDP, + = deficit)	Personal saving rate (%)	0.576
Deficit (% GDP, + = deficit)	Federal receipts (% of GDP)	-0.581
Old-age dependency ratio	German real 10y rate	-0.877
Old-age dependency ratio	Unemployment rate (%)	-0.333

Old-age dependency ratio	Personal saving rate (%)	0.077
Old-age dependency ratio	Federal receipts (% of GDP)	0.034
German real 10y rate	Unemployment rate (%)	0.263
German real 10y rate	Personal saving rate (%)	0.191
German real 10y rate	Federal receipts (% of GDP)	0.089
Unemployment rate (%)	Personal saving rate (%)	0.485
Unemployment rate (%)	Federal receipts (% of GDP)	-0.572
Personal saving rate (%)	Federal receipts (% of GDP)	-0.143

Notes: Entries are Pearson correlations from corr_matrix.xlsx. Only unique pairs are reported (upper triangle).

Table A4 Descriptive Statistics core-controls specification variables (1960-2023)

Variable	Mean	Std. dev.	Min	Max
U.S. real 10y rate	2.045	2.480	-5.048	8.138
Deficit (% GDP, + = deficit)	3.018	2.962	-2.305	14.669
Old-age dependency ratio	18.320	2.786	14.743	26.829
Deficit $\times$ Old-age dependency ratio	59.934	67.599	-42.135	359.746
German real 10y rate	2.619	2.387	-5.362	6.296
Unemployment rate (%)	5.917	1.618	3.490	9.708
Personal saving rate (%)	8.451	3.244	2.250	15.092
Federal receipts (% of GDP)	17.852	1.006	15.485	20.173

Notes: Descriptive statistics computed from final_analysis_dataset.dta. Annual observations, 1960–2023. All variables have $N = 64$. Variables correspond to the core-controls specification (Table 2, column 4).

